MIL-T-45696 (Ord)
15 June 1961
SUPERSEDING
FA(FC)-PD-536
11 September 1959

MILITARY SPECIFICATION

TEST PANEL: 8643947

1. SCOPE

1.1 This specification covers one type of test panel used in the Gun Direction Computers: M15 and M15C which are part of Field Artillery Fire Control Systems: M35 and M35C. The test panel performs six separate functions as described in 6.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

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DRAWINGS

Ordnance Corps

F8643947	Test Panel
F8643948	Electrical Schematic, Test Panel

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one test panel, representative of production, shall be made available to the Government to determine that the test panel meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The test panel shall be manufactured in accordance with Drawing F8643947, "Test Panel", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential.- The design and construction of the test panel shall insure that all external parts, exclusive of electrical connections, are at ground potential at all times.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926.

(a) Order of precedence
(b) Protective finished surface
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

3.5 Environmental.

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3.5.1 Vibration.- The test panel shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.03 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the test panel shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperatures.- The test panel shall meet all the requirements of 3.6 and 3.7 when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.5.3 Storage temperatures.- The test panel shall show no evidence of physical failure and shall meet the requirements of 3.6 and 3.7 at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 65 degrees Fahrenheit.

3.6 Circuitry.- Circuitry shall be in accordance with electrical schematic, Drawing F8643948.

3.7 Performance.

3.7.1 Muzzle velocity section.

3.7.1.1 Torque.- The torque required for rotation of the knobs shall be within the range of 4 to 10 ounce inches at room temperature (60 to 90 degrees Fahrenheit) and at the temperatures specified in 3.5.2.

3.7.1.2 Potentiometer shafts.- The potentiometer shafts shall rotate in a clockwise direction for increasing values of muzzle velocity.

3.7.1.3 Linearity of potentiometers.- When the muzzle velocity potentiometers are energized with plus and minus 2.5 volts 400 cps, each output voltage for potentiometers 1 through 7 shall be linearly proportional to its respective dial setting and shall not exceed the tolerance specified in Table I for the dial settings specified under a no load condition.

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TABLE I

Dial Settings (feet per second)	Output Voltages (volts)	Tolerances (millivolts)
-40	-2.000	±25
-10	-0.500	±25
0	0.000	±10
+10	+0.500	±25
+40	+2.000	±25

3.7.2 Test function section.- Unless otherwise specified, all voltages used to determine the performance requirements are those shown on Drawing F8643948.

3.7.2.1 Normal.- When all of the test switches are positioned to NORMAL, all test circuitry shall be bypassed and the normal inputs to the computer circuits shall be transmitted unchanged.

3.7.2.2 Zero effects.- When those switches with an indicated zero position are positioned to zero, the selected output circuits for those switches shall be referred to signal ground; and zero values of the selected functions shall be transmitted to the computer.

3.7.2.3 Fixed effects.

3.7.2.3.1 East-west (EW), north-south (NS), air temperature (AT) and air density (AD) resistors.- The output voltage shall be within the respective tolerance specified when the switch is positioned in accordance with Table II and the EW, NS, AT and AD test resistors are energized with plus and minus 2.5 volts, 400 cps. The output circuits shall be loaded with 250K-Ohms plus or minus 0.5 percent.

TABLE II

Switch Positions	Output Voltage (volts)	Tolerances (millivolts)
EW -	-1.250	±25
EW +	+1.250	±25
NS -	+1.250	±25
NS +	-1.250	±25
AT -	-1.756	±25
AT +	+1.756	±25
AD -	-1.000	±25
AD +	+1.000	±25

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3.7.2.3.2 Geometry (geom) resistors.- The output voltages shall be within the respective tolerances specified when the switch is positioned in accordance with Table III and the geometry test resistors energized with plus or minus 15 volts, 400 cps. The output circuit shall be loaded with 250K-ohms plus or minus 0.5 percent.

TABLE III

Switch Position	Output Voltage and Designation (volts)		Tolerance (millivolts)
Geom 1-4	-2.419	△E Selected	±25
Geom 1-4	-1.818	△N Selected	±25
Geom 1-4	-0.925	△H Selected	±25
Geom 5-7	+4.855	△E Selected	±25
Geom 5-7	+3.654	△N Selected	±25
Geom 5-7	-0.925	△H Selected	±25

3.7.3 Line number test switch.- When the line number test switch is positioned to LINE NO. CUTOUT BYPASS, a plus 28 volts, dc, unregulated, shall be transmitted to the computer at the lead designated +28V DC UNREG LN BYPASS. When the switch is positioned to NORMAL this circuit shall open.

3.7.4 Test indicator signal.- When any switch except GEOM and LINE NO. is in some position other than normal, a plus 28 volts, dc, unregulated, shall be transmitted to the computer by means of the lead marked +28V DC for test indicator on Drawing F8643948.

3.7.5 Air temperature and range wind indicators.- When the following signals are introduced at the points respectively identified on Drawing F8643948 the respective indicator shall be illuminated.

Signal	Indicator Light
(a) +28V DC UNREG + Wr	(Wr) +
(b) +28V DC UNREG - & 0 Wr	(Wr) - & 0
(c) +28V DC UNREG + AT	(AT) +
(d) +28V DC UNREG - & 0 AT	(AT) - & 0

3.7.6 Illumination knob.- When a test lamp is connected in series with a variable resistor and 28 volts, dc applied through the circuit, the manipulation of the illumination knob controlling the wiper arm of the resistor shall vary the intensity of illumination.

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3.7.7 Elapsed time meter.- The elapsed time meter shall be capable of operating from 115 plus or minus 5 volts, 400 cps plus or minus 5 percent.

3.8 Alinement and calibration.- Alinement and calibration of all test panels shall be performed using the same power source to insure uniformity of product. The reference voltages supplied shall be from a power source maintained as a standard.

3.9 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced specifications, to the extent specified herein, together with the provisions of this section are applicable to inspection performed by the contractor and Government verification inspectors.

4.1.1 Contractor's inspection.- Contractor's inspection system and Government verification of contractor's system shall be in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one test panel shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

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4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the test panel. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table IV.

TABLE IV - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values

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within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Accuracy requirements for special testing devices.

4.6.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion as specified. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in the specified positions.

4.6.2.2 Device to test torque.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to twenty-five ounce-inches, the accuracy of which shall be within plus or minus one quarter pound-inches throughout the entire zone of excursion. The device shall be equipped with an adaptor to accommodate the knobs.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables V and VI shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE V - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-a of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't. Par.	QA. Par.	Test Method
101. Post vibration	3.5.1	4.7.2	Device, Special Testing (DST)
102. Operating temperatures	3.5.2	4.7.3	DST

MINOR: NONE DEFINED

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TABLE VI - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A
of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't. Par.	QA. PAR.	Test Method
101.	Torque	3.7.1.1	4.7.5	Device, Special Testing (DST)
102.	Potentiometer shaft	3.7.1.2	4.7.6	Visual-Tactile
103.	Linearity of potentiometers	3.7.1.3	4.7.7	Standard Measuring Equipment (SME)
104.	Normal switch positions	3.7.2.1	4.7.8	SME
105.	Zero switch positions	3.7.2.2	4.7.9	SME
106.	(EW), (NS), (AT), (AD) resistors	3.7.2.3.1	4.7.10	SME
107.	Geometry resistors	3.7.2.3.2	4.7.11	SME
108.	Line number test switch	3.7.3	4.7.12	SME
109.	Test indicator signal	3.7.4	4.7.13	SME
110.	(AT) and (RW) indicator	3.7.5	4.7.14	Visual-Tactile
111.	Illumination knob	3.7.6	4.7.15	Visual-Tactile
112.	Elapsed time meter	3.7.7	4.7.16	SME
113.	Circuitry	3.6	4.7.17	SME
114.	Identification marking	3.9	4.7.18	Visual

MINOR: NONE DEFINED

4.7.2 Vibration.- This test shall be conducted in accordance with Method 201 of Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements prior to or during vibration are not required for final acceptance. Subsequent to vibration, the test panel shall be subjected to a visual-tactile examination for damage. The unit shall then be subjected to all tests in this specification.

4.7.3 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the test panel is stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.7.5 to 4.7.17 inclusive and shall meet the requirements specified.

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4.7.4 Storage temperature.- This test shall be conducted in accordance with Procedure II, condition "B" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the test panel shall be subjected to tests specified in 4.7.5 to 4.7.17 inclusive and shall meet the requirements specified. This test shall be performed on the production model only.

4.7.5 Torque.- This test shall be performed with the aide of a special testing device, affixed to the knobs, that shall be capable of applying torque to the knob within the range specified. The torque required for rotation of the knobs shall be within the values specified.

4.7.6 Potentiometer shafts.- Potentiometer shafts shall increase values of muzzle velocity when rotated in a clockwise direction.

4.7.7 Linearity of potentiometers.- The muzzle velocity potentiometers shall be energized as directed and the dial settings adjusted to the settings listed in Table I. The measured output voltages shall coincide with the values listed, within the allowable tolerances.

4.7.8 Normal switch positions.- The measured transmission of the normal inputs to the computer circuits shall be unchanged with the test switches positioned to NORMAL.

4.7.9 Zero switch positions.- When the switches with an indicated zero position are so positioned, the effects on the selected output circuits and zero values of the selected functions shall comply with the requirements.

4.7.10 EW, NS, AT and AD resistors.- The output voltages shall be measured for compliance with the requirements in Table II, when the resistor switches are adjusted as indicated, the resistors energized as directed, and the circuits loaded with the required resistance.

4.7.11 Geometry resistors.- The geometry resistors shall be energized as directed and the output voltages measured for compliance with the requirements in Table III.

4.7.12 Line number test switch.- The computer lead designated +28V DC UNREG LN BYPASS shall be checked for compliance with the requirements when the line number test switch is positioned as designated.

4.7.13 Test indicator signal.- The lead marked +28V DC shall be checked for the required voltage when any switch other than GEOM and LINE NO. is in a position other than normal.

4.7.14 AT and RW indicators.- The illumination of the AT and RW indicators shall be checked for compliance with requirements.

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4.7.15 Illumination knob.- The variable resistor controlling illumination shall be energized and checked for compliance with requirements specified.

4.7.16 Elapsed time meter.- The accuracy of the elapsed time meter shall comply with the requirements when energized as directed.

4.7.17 Circuitry.- An electrical point-to-point measuring check shall be made using standard measuring equipment to assure that the values of the circuits conform to the electrical schematic.

4.7.18 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The test panel shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619/P8537. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The test panel is a plug-in subassembly of Gun Direction Computers: M15 and M15C. The unit is intended to perform six functions:

(a) When associated with the muzzle velocity unit, it will transmit analog voltages of△MV (changes in muzzle velocity), charges 1 through 7 to the computer.
(b) Establish various test conditions through manually positioned rotary and toggle switches.
(c) Controls computer illumination.
(d) Records accumulative computer operating time.
(e) Transmits operating and reference voltages to a field test set.
(f) Provides for mounting of the H/R and Line Unit and a window for viewing the dial gears of the H/R and Line Unit.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

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6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 Special testing device (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

6.4.4 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

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